



 **Time:** 5 minutes

 **Outcome:** Apply the natural-rate hypothesis and vertical long-run Phillips curve.

Difficulty:  Intermediate

Long-Run Phillips Curve



THE CORE IDEA

The long-run Phillips curve is vertical at the natural unemployment rate because expected inflation eventually adjusts. A change in labor-market structure can shift the vertical LRPC to a new natural rate. Expected inflation adjusts, the SRPC shifts, and unemployment returns toward its natural rate. The long-run Phillips curve is vertical at the natural rate of unemployment.



HOW TO RECOGNIZE IT

- The long-run Phillips curve is vertical at the natural rate of unemployment.
- Expected inflation rises from 2 percent to 5 percent while unemployment remains at its natural rate.
- The LRPC is vertical at the natural rate because expected inflation adjusts in the long run.
- A national job-matching system reduces the time workers spend searching for suitable positions.



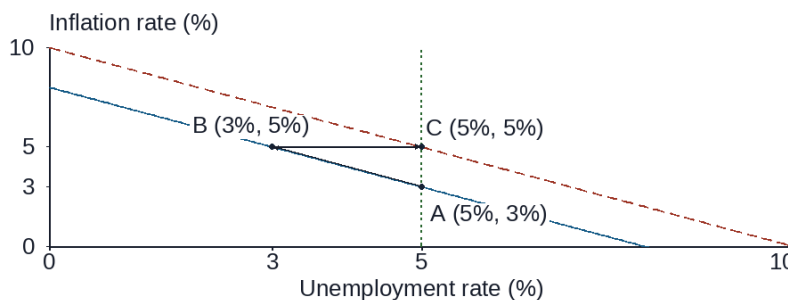
WATCH OUT

Do not treat a higher expected inflation rate as a permanent way to lower unemployment. Along the long-run Phillips curve, different inflation rates can coexist with the same natural rate of unemployment.



WORKED EXAMPLE: RETURNING TO THE NATURAL RATE

— SRPC0 - - - SRPC1 ... LRPC



The economy starts at A: 5% unemployment and 3% inflation. A demand expansion moves it along SRPC0 to B, at 3% unemployment and 5% inflation. As expected inflation rises, SRPC shifts up to SRPC1. The final point C is on LRPC at 5% unemployment and 5% inflation. Unemployment returns to its natural rate; there is no permanent inflation-unemployment tradeoff.



CHECK YOURSELF

Why does higher inflation not permanently lower unemployment in the natural-rate model?



READY?

Return to the game and master this concept.